Tertiary Infotech Academy Pte Ltd

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LEARNER GUIDE

For

Design Thinking Course for Businesses

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Conducted by

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
14.0	1 January 2024	Previous release of the Learner Guide, issued under the superseded course reference.	Dr Alfred Ang
15.0	10 August 2026	Re-issued under WSQ course code TGS-2026064719. Content carried over from the v14 master trainer deck and restructured to the current Tertiary Infotech WSQ house standard. Added 7 step-by-step hands-on activities built on the browser-based ed-tools, expanded topic notes, an assessment-preparation section, a glossary and references.	Dr Alfred Ang

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Introduction

This Learner Guide accompanies the WSQ course Design Thinking Course for Businesses (TGS-2026064719), conducted by Tertiary Infotech Academy Pte Ltd. It provides step-by-step instructions for all 7 hands-on activities, organised by the four course topics, and is aligned to the Skills Framework for Design TSC "Design Thinking Practices" (DSN-ACE-3014-1.1).

Design thinking is a human-centred approach to problem solving that aims to improve people's experience. It is as much a mindset as a process. This guide is written so you can follow every activity in class and repeat it afterwards at your own workplace — the activities use free, browser-based ed-tools that stay available to you after the course.

Use this guide alongside the course slides and the activity files in the labs/ folder. The slide deck is deliberately visual and does not repeat the step-by-step instructions — those live here.

Course Learning Outcomes

- LO1: Understand key concepts of design thinking to communicate the design outcomes
- LO2: Apply design thinking to generate new ideas for the organization
- LO3: Uncover opportunities of design thinking
- LO4: Implement plans to embed various stages of design thinking across the organization
- LO5: Execute the design concept through prototypes
- LO6: Utilize metrics to measure the outcomes of design ideas and solutions

Skills Framework Alignment

TSC Title: Design Thinking Practices · TSC Code: DSN-ACE-3014-1.1

Abilities

- A1: Apply design thinking methodologies to define design problems and generate new ideas for the organization
- A2: Uncover opportunities for applying design thinking across the organization
- A3: Utilize metrics to benchmark and measure outcomes of design ideas and solutions
- A4: Implement plans to embed design thinking across the organization
- A5: Facilitate the development of design concepts through prototypes and visual tools
- A6: Communicate design outcomes and their value to stakeholders

Knowledge

- K1: Key concepts and principles of design thinking
- K2: Importance and business value of design thinking
- K3: Traits and mindset of a design thinker
- K4: Action phases of design thinking and the activities within each phase
- K5: Use cases and applications of design thinking across industries
- K6: Approaches to applying design thinking within an organization
- K7: Methodologies, visual tools and metrics used at each action phase

Before You Start — Your Ed-Tools

Every activity in this course runs in your web browser. Nothing needs to be installed, and every tool below remains free to use after the course, so you can run the same activities with your own team.

- Design Thinking Toolkit — https://alfredang.github.io/designthinking/ — Empathy Map, Persona, Journey Map, POV, HMW, Brainstorm, Prototype and Test canvases — the primary tool for this course.
- RACI Matrix — https://alfredang.github.io/raci/ — Assign Responsible / Accountable / Consulted / Informed roles so a design idea has a clear owner when it moves to delivery.
- Scrum Planner — https://alfredang.github.io/scrum/ — Turn validated design concepts into a prioritised product backlog and sprint plan.
- Digital Transformation Canvas — https://alfredang.github.io/digitaltransformation/ — Position a design initiative within the organisation's wider transformation roadmap.
- Business Continuity Management — https://alfredang.github.io/bcm/ — Stress-test a design solution for operational risk and continuity before rollout.

Also used in class

- draw.chat — a free collaborative whiteboard for quick sketching; share the board URL with the class.
- padlet.com — a shared wall used to post your work so the whole class can see and comment on it.
- Paper, pens and sticky notes — the fastest prototyping tools ever invented. Do not underestimate them.

Conventions used in every activity

- Each activity states its objective, what you will produce, the steps, and how to check you are done.
- Timings shown are the in-class timings; take longer when you repeat the activity at work.
- Where an activity says 'interview your partner', use their exact words — verbatim quotes carry the insight.

- Do not skip the 'You're done when' check; it is the same evidence the trainer looks for in the practical assessment.

How to Download Your Course Material

1. Go to the LMS/TMS portal at https://lms-tms.tertiaryinfotech.com and sign in with the e-mail address you registered with.
2. Open My Courses and select Design Thinking Course for Businesses.
3. Download the Trainer Slides (PPT), Learner Slides (PDF), this Learner Guide (PDF) and the Lesson Plan (PDF).
4. Keep the slides and this guide open during the assessment — it is open book.
5. Complete the mandatory TRAQOM survey on the same portal at the end of the course.

Topic 01 — Key Concepts & Principles of Design Thinking

Concept · Importance · Traits of a design thinker · Applying design thinking for the organisation

Skills mapped: A6, K1, K2, K3, A1, K6

Key concepts

- What design thinking is — A human-centred problem-solving approach that aims to improve people's experience — as much a mindset as a process.
- Why it matters — Design-led organisations report higher market share, stronger competitive advantage and more loyal customers.
- Not just for designers — Anyone can think like a designer to create products and services that improve the customer experience.
- Analytical vs creative — Design thinking deliberately pairs left-brain analysis with right-brain creativity.
- The design thinker's mindset — Think users first, ask the right questions, sketch, commit to explore, prototype to test.
- Desirability · Viability · Feasibility — Organisational design thinking lives at the intersection of user desire, business viability and technical feasibility.

What design thinking is

Design thinking is a problem-solving approach that aims to improve people's experience. It is human-centred: it starts from a deep understanding of customers' needs and wants, encourages creative consideration of a wide array of innovative solutions, and is as much a mindset as a process.

Why it matters to a business

Research by the Design Management Institute found design-led organisations reported 41% higher market share, 46% greater competitive advantage overall, 50% more loyal customers, and 70% saying their digital experiences beat competitors. Design thinking offers a problem-solving roadmap, unlocks innovative ideas, solves genuine human problems, creates behavioural change and increases customer satisfaction.

It is not just for designers

Design thinking is a mindset anyone can apply. It means thinking like a designer so that you can create a new product or service that improves your customer's or user's experience — regardless of your job title.

Analytical and creative thinking together

Left-brain thinking is analytical, rational, objective, focused on the present and past, on facts, order and planning. Right-brain thinking is creative, holistic, subjective, focused on the present and future, on feelings, space and spontaneity. Design thinking deliberately uses both.

The mindset shift

Traditional thinkers say "we have this problem, let's get in a room and brainstorm solutions", "our competitors just launched X, how can we do X quickly?" or "we have this technology, what can we use it for?". Design thinking shifts focus from a business-centric engineering solution to a customer-centric one: think users first, ask the right questions, believe you can sketch, commit to explore, and prototype to test and evaluate. It is also a growth mindset — ability is a starting point, not a ceiling.

Applying it to an organisation

The problems an enterprise faces are much bigger than designing a vase. Organisational design thinking happens at the intersection of three lenses: desirability for customers, viability at the business level, and feasibility for technology. A solution must satisfy all three.

Activity 1 — Design a Vase — Reframing the Brief

Objective: LO1 / K1, K3 — experience how a narrowly framed brief limits creativity, and how reframing around the user unlocks it.

Goal: A two-round warm-up that shows the difference between designing an object and designing an experience. In round 1 you design a vase. In round 2 you re-design against a human-centred brief and compare how much wider the solution space becomes.

What you'll produce

Two sketched concepts and a written reflection on how the framing of a brief drives creativity. (Ed-tools: Design Thinking Toolkit (alfredang.github.io/designthinking), draw.chat, Padlet.)

Step-by-step

1. Open the free whiteboard at https://draw.chat/ and create a new board.
2. Round 1 — you have 2 minutes. Sketch a design for a vase. Do not overthink it.
3. Share your board URL with the class on https://padlet.com/ and review what everyone else drew.
4. Discuss: how similar were the designs? Note that almost every vase is a cylinder that holds water.
5. Round 2 — re-read the brief as 'Design a better way for someone to enjoy flowers in their home.'
6. Ask the user questions first: Why would someone buy a vase? What purpose does it serve? Is a vase the only way to enjoy flowers?
7. Sketch a second concept against the reframed brief and post it to Padlet alongside the first.
8. Write a 3-line reflection: which prompt produced more creativity, and why the narrow prompt limited it.

You're done when

You have two visibly different concepts on your board, and you can state in one sentence why the second brief produced a wider range of ideas than the first.

Note: The full activity brief is in labs/lab-01-*.md. Keep your output — it is the evidence used in the Practical Performance assessment.

Activity 2 — Design Thinking Readiness — Mindset & Culture Self-Assessment

Objective: LO1, LO4 / K2, K3, K6 — assess your organisation's design-thinking mindset and identify what must change.

Goal: Benchmark your own team against the traits of a design thinker and the three lenses of organisational design thinking — desirability, viability and feasibility — then identify the single biggest barrier to embedding design thinking where you work.

What you'll produce

A completed readiness scorecard with a prioritised list of cultural barriers and first actions. (Ed-tools: Design Thinking Toolkit (alfredang.github.io/designthinking), Digital Transformation Canvas (alfredang.github.io/digitaltransformation).)

Step-by-step

1. Open the Digital Transformation Canvas at https://alfredang.github.io/digitaltransformation/.

2. Score your team 1-5 on each design-thinker trait: think users first, ask the right questions, sketch, commit to explore, prototype to test.
3. For each of the three lenses — desirability (customers), viability (business), feasibility (technology) — note where your current projects are strongest and weakest.
4. Contrast the traditional mindset ('we have this technology, what can we use it for?') with the design mindset ('what does the user actually need?') using a real example from your workplace.
5. Identify the ONE cultural barrier that most blocks design thinking in your organisation.
6. Write one concrete first action you could take in the next 30 days to reduce that barrier.
7. Export or screenshot your canvas and share it with the class for discussion.

You're done when

Your scorecard is complete across all three lenses, and you have named one barrier and one specific, time-bound action to address it.

Note: The full activity brief is in labs/lab-02-*.md. Keep your output — it is the evidence used in the Practical Performance assessment.

Topic 02 — Applications of Design Thinking

Use cases across industries · Uncovering opportunities for applying design thinking

Skills mapped: A2, K5

Key concepts

- Airbnb — Built a culture of experimentation and design-led iteration to move from failing start-up to a billion-dollar business.
- IBM — Applies design thinking at scale to complex teams, problems and organisations through IBM Design Thinking.
- Bank of America — Partnered with IDEO to research real savers, producing the 'Keep the Change' account.
- Uber Eats — Designs its food-delivery service around observed customer, courier and restaurant journeys.
- Healthcare — Stanford's Hasso Plattner Institute of Design redesigned the emergency-room patient experience.
- Public & social sector — Clean Team (Ghana in-home toilets), Golden Gate Regional Center and The Good Kitchen redesigned public services.

Airbnb

Design thinking is part of Airbnb's success. In particular they built a culture of experimentation — going out to meet hosts, re-photographing listings by hand, and iterating relentlessly — which took them from a failing start-up to a billion-dollar business.

IBM

IBM applies design thinking to complex teams, problems and organisations through its own IBM Design Thinking framework, restructuring how large product teams work.

Bank of America

Bank of America partnered with design consultancy IDEO in 2004 to understand how to get more people to open bank accounts. Observing real people revealed that savers rounded up their spending — producing the 'Keep the Change' account.

Uber Eats

The Uber Eats team designs its food-delivery service with a design-thinking mindset, studying the journeys of customers, couriers and restaurants in the field.

Healthcare — Hasso Plattner Institute

The Hasso Plattner Institute of Design at Stanford explored design-thinking approaches to improve the patient experience in the emergency room.

Public and social sector

Clean Team applied design thinking to provide in-home toilets for Ghana's urban poor. The Golden Gate Regional Center redesigned services and financial support for people with developmental disabilities. In Denmark, Hatch & Bloom redesigned a municipal meal service into The Good Kitchen, giving elderly residents more quality, flexibility and choice.

Uncovering your own opportunities

Look for friction — where customers or staff visibly struggle, complain or invent workarounds. A workaround is a user telling you the design is wrong. Follow repeated complaints and support tickets; they are free user research. Find where people abandon a journey and ask what happened just before. Then rank the opportunities by impact and effort before committing a team.

Activity 3 — Uncover Design Thinking Opportunities in Your Business

Objective: LO3 / A2, K5 — analyse published use cases and uncover opportunities for applying design thinking in your own organisation.

Goal: Study how Airbnb, IBM, Bank of America, Uber Eats and the public-sector cases applied design thinking, extract the transferable pattern from each, then map three concrete opportunities in your own business and rank them by impact and effort.

What you'll produce

An opportunity map with three ranked design-thinking opportunities for your own organisation.   (Ed-tools: Design Thinking Toolkit (alfredang.github.io/designthinking), Digital Transformation Canvas (alfredang.github.io/digitaltransformation), Padlet.)

Step-by-step

1. Review the case studies covered in class: Airbnb, IBM, Bank of America, Uber Eats, Clean Team, Golden Gate Regional Center and The Good Kitchen.
2. For any two cases, write down: who was the user, what was the insight, and what changed as a result.
3. Open https://alfredang.github.io/designthinking/ and start a new opportunity map.
4. List three processes, products or services in your organisation where customers or staff visibly struggle.
5. For each, write the user group affected and the symptom you can actually observe (not the solution).
6. Plot the three opportunities on an impact-versus-effort grid and pick the one worth running a design sprint on.
7. Post your top opportunity to https://padlet.com/ and explain in two lines why it is the right place to start.

You're done when

You can name three design-thinking opportunities in your own organisation, each with an identified user group and observable symptom, and justify which one you would tackle first.

Note: The full activity brief is in labs/lab-03-*.md. Keep your output — it is the evidence used in the Practical Performance assessment.

Topic 03 — Action Phases of Design Thinking

The Stanford d.school five-stage model · Empathize · Define · Ideate · Prototype · Test

Skills mapped: A4, K4, A1

Key concepts

- Empathize — Understand the user's experience, situation and emotion by observing, engaging and immersing — without judging.
- Define — Synthesise the findings into a meaningful, actionable problem statement (user + need + insight).
- Ideate — Go wide: generate a large variety and quantity of ideas before narrowing to the strongest few.
- Prototype — Build to think — a cheap, fast, low-fidelity artefact that makes the idea tangible.
- Test — Put the prototype in users' hands, gather feedback, and refine or reframe.
- Iterative, not linear — The five phases loop — insights from Test routinely send you back to Empathize or Define.

The five-stage model

The five-stage design thinking model was proposed by the Hasso-Plattner Institute of Design at Stanford University (d.school): Empathize, Define, Ideate, Prototype and Test. The phases form a loop, not a straight line — insights from testing routinely send you back to empathising or redefining the problem.

Phase 1 — Empathize

Design thinking cannot begin without a deep understanding of the people you are designing for. Objective: to understand the experience, situation and emotion of your user. Observe (view users and their behaviour in the context of their lives, without judging), Engage (interact in conversations and interviews; ask why) and Immerse (experience what your user experiences). Assume a beginner's mindset — consciously set aside your own assumptions. The classic example is the MRI scanner: immersion revealed that an MRI room is a frightening experience for a child, and the redesign turned the scan into an adventure story.

Phase 2 — Define

An integral part of the process is defining a meaningful, actionable problem statement. Objective: process and synthesise your findings to form a problem statement you will address. Understand the User (the persona), the Needs (always expressed as verbs) and the Insights (what you discovered). A good problem statement is human-centred, broad enough to allow creativity, and narrow enough to be manageable. Methods include clustering and bundling ideas and facts, empathy mapping, POV statements, 'How Might We' questions and why-how laddering.

Phase 3 — Ideate

Ideation is where you concentrate on idea generation — going wide in terms of concepts and outcomes. Objective: translate problems into solutions by exploring a wide variety and large quantity of ideas, going beyond the obvious. It relies on creativity (combining rational thought with imagination), group synergy (building on each other's ideas) and the separation of divergent from convergent thinking. Active facilitation matters: create a curious, courageous and concentrated atmosphere.

Phase 4 — Prototype

Prototyping produces an early, inexpensive, scaled-down version of the product to reveal problems with the current design. Objective: build to think — a simple, cheap, fast way to shape ideas so you can experience and interact with them. A prototype can take any physical form: a wall of post-it notes, a role-play, a space, an object, an interface or a storyboard. Low-fidelity prototypes use basic models and simple materials; high-fidelity prototypes are closer to the finished product. Guidelines: just start building, don't spend too much time, remember what you are testing for, and build with the user in mind.

Phase 5 — Test

Testing generates user feedback on your prototypes and deepens your understanding of your users. Objective: ask for feedback, learn about your user, reframe your POV and refine the prototype. Show (let people use it and listen), Create experiences (let them describe how it feels) and Compare (let users test multiple prototypes to reveal latent needs). Test in a natural setting wherever possible. Negative feedback is valuable: if users struggle, revisit your solutions and look for problems you had not considered.

Activity 4 — The Wallet Project — A Full Design Thinking Cycle

Objective: LO2, LO4, LO5 / A1, A4, K4 — run all five action phases end to end in a single compressed cycle.

Goal: The classic Stanford d.school immersive exercise. Working in pairs, you take a partner through all five action phases — Empathize, Define, Ideate, Prototype and Test — to redesign their wallet, or more precisely the experience of carrying what matters to them. It is deliberately fast so you feel the rhythm of the whole cycle.

What you'll produce

A tested paper prototype of a redesigned wallet, plus your partner's feedback and a refined concept. (Ed-tools: Design Thinking Toolkit (alfredang.github.io/designthinking), draw.chat, Padlet.)

Step-by-step

1. Pair up. Decide who is the designer first — you will swap roles later.
2. EMPATHIZE (6 min) — interview your partner about their wallet. Ask what they carry, when it annoys them, and why. Ask 'why' at least three times to get past the surface answer.
3. Sketch what you heard, then interview again to dig into the emotion behind the annoyance, not just the function.

4. DEFINE (3 min) — write a POV problem statement: [USER] needs a way to [NEED] because [INSIGHT]. Use your partner's own words for the insight.
5. Check the statement your partner agrees it is true. If they hesitate, the insight is wrong — go back and ask again.
6. IDEATE (5 min) — sketch at least five radically different solutions. Go for volume; include at least one deliberately absurd idea.
7. Show the sketches to your partner and let them pick the one they most want to try.
8. PROTOTYPE (7 min) — build a physical or drawn prototype of the chosen idea using paper, sticky notes, or https://draw.chat/. It must be something they can hold or point at.
9. TEST (5 min) — hand the prototype to your partner and stay silent. Let them use it and narrate what they think it does. Record what surprises you.
10. Capture their feedback in two columns: what worked, and what confused them. Do not defend your design.
11. Refine the prototype once based on that feedback, then swap roles and repeat the whole cycle.
12. Post a photo of your final prototype to https://padlet.com/ with your one-line POV statement.

You're done when

You have completed all five phases, your prototype changed as a direct result of your partner's test feedback, and you can state the POV insight in your partner's own words.

Note: The full activity brief is in labs/lab-04-*.md. Keep your output — it is the evidence used in the Practical Performance assessment.

Topic 04 — Methodologies and Visual Tools for Design Thinking

Empathize · Define · Ideate · Prototype · Test methods, tools and metrics

Skills mapped: A5, K7, A3

Key concepts

- Empathize tools — Empathy Map (Says / Thinks / Does / Feels), Persona Mapping, Journey Mapping, interviews and bodystorming.
- Define tools — Point of View (POV) statements, 'How Might We' questions, Why-How laddering and the Business Model Canvas.
- Ideate tools — Brainstorming, mind mapping, sketchstorm, analogies and gamestorming; then dot voting or a Now-Wow-How matrix to select.

- Prototype tools — Low- and high-fidelity prototypes: paper sketches, storyboards, wireframes, cardboard and clay models.
- Test tools — User test scripts, feedback grids, evaluation matrices and A/B comparison of alternative prototypes.
- Measuring outcomes — Traditional KPIs, customer feedback (CSAT/NPS), design-thinking activity metrics and business-value/novelty mapping.

Methods by action phase

Empathize produces empathy maps, persona maps, lists of user feedback and identified problems. Define produces a design brief (POV and HMW), stakeholder maps, context maps, customer journeys and opportunity maps. Ideate produces ideas and concepts, sketches, prioritisation maps, affinity maps and idea evaluations. Prototype produces physical prototypes, wireframes and storyboards. Test produces user feedback, observations, evaluation maps and proposed refinements.

The empathy map

Empathy maps are split into four quadrants — Says, Thinks, Does and Feels — with the user or persona in the middle. SAYS contains verbatim quotes from the interview. THINKS captures what the user was thinking but did not say out loud. DOES records observable actions. FEELS records the emotional state and its intensity. Empathy maps give a glance into who a user is as a whole; they are not chronological. Users are complex, so contradictions between quadrants are normal and extremely beneficial — a positive quote next to a negative emotion is exactly where the real insight hides.

Persona and journey mapping

Persona mapping identifies who your clients are and how they make decisions, then applies that to more effective strategies. Journey mapping charts every touchpoint over time — before, during and after the core interaction — exposing where the experience breaks down and which moments disproportionately shape the whole experience.

Point of View (POV)

POV = USER + NEED + INSIGHT. The template is: [USER] needs a way to [USER'S NEED] because [INSIGHT]. For example: "A busy working mom needs to be able to bring plastic bags with her when she's not at home, because she buys the equivalent of three dog-poop bags a day cleaning up after her dog." Or: "A teacher needs to clean and store his used sandwich bags at school, because every morning he uses a new bag and often discards them rather than washing them." Developing a strong POV is challenging but shapes the entire future of the project. Always record the underlying assumptions so they can be tested. The Business Model Canvas is a complementary Define tool for reflecting systematically on the business model segment by segment.

Ideation methods

There are hundreds: brainstorm, mind map, sketch or sketchstorm, storyboard, analogies, provocation, movement, bodystorm, gamestorming, cheatstorm, crowdstorm, co-creation workshops, prototyping and the creative pause. Brainstorming leverages the collective thinking of the group by engaging, listening and building on others' ideas. Rules: set a mission, set up the space, limit the time, don't hang on to any one idea too tightly, defer judgement and go for volume — 10 ideas are better than 3, and 200 are better than 50. Once the session is complete, collect, categorise, refine and narrow down using post-it or dot voting, the four categories method, bingo selection, idea affinity maps or a Now-Wow-How matrix.

Prototyping and testing in practice

Prototype as if you know you're right, but test as if you know you're wrong. Ask what the cheapest, quickest test is that would prove or disprove your assumption. Give the test a name and a description, decide where and when you will run it, and state what metrics will measure success — pre-orders, votes, completion, smiles. Write down which assumptions this particular prototype tests and the minimum result that would count as success. When planning the test, let users compare alternatives, show rather than tell, and ask them to talk through their experience.

Measuring the outcome

Measure design thinking across several categories: traditional KPIs (increased sales, ROI per project and other financial measures); customer feedback (customer satisfaction, net promoter score, campaign response, usability metrics); design-thinking activity (number of projects, people trained, coaches trained); and quick results (concepts finished, projects launched, funded or in development). Move beyond execution-oriented metrics to track creative behaviours. The three main drivers leading companies to pursue design thinking are: to better understand customers or end users, to protect business share from disruption and start-ups, and to develop more innovative methods and team dynamics. Plot concepts on a business-value versus novelty grid — valuable and novel is the target quadrant.

Activity 5 — Empathy Map & Persona — The Gift-Giving Experience

Objective: LO2 / A1, A5, K7 — build an empathy map and persona to capture what a user says, thinks, does and feels.

Goal: Redesign the gift-giving experience for a partner. You start by building empathy: interview them about the last gift they gave, then plot what they said, thought, did and felt into a four-quadrant empathy map and distil it into a persona.

What you'll produce

A completed four-quadrant empathy map and a one-page persona for your partner. (Ed-tools: Design Thinking Toolkit — Empathy Map & Persona canvases (alfredang.github.io/designthinking).)

Step-by-step

1. Open https://alfredang.github.io/designthinking/ and select the Empathy Map canvas.
2. Interview your partner for 5 minutes about the last gift they gave: what they did, how they chose it, and how it felt.
3. SAYS — record direct quotes, verbatim. Do not paraphrase; the exact words carry the insight.
4. THINKS — record what they were thinking but did not say out loud, such as worries about cost or judgement.
5. DOES — record the observable actions: where they shopped, how long it took, who they asked.
6. FEELS — record the emotions and their intensity: anxious, rushed, proud, relieved.
7. Look for contradictions between quadrants — a positive quote next to a negative feeling is where the real insight hides.
8. Switch to the Persona canvas and distil the map into a named persona with goals, frustrations and one defining quote.
9. Share your empathy map with your partner and ask them to correct anything you got wrong.

You're done when

All four quadrants are populated with your partner's own words, you have identified at least one contradiction between quadrants, and your partner confirms the persona sounds like them.

Note: The full activity brief is in labs/lab-05-*.md. Keep your output — it is the evidence used in the Practical Performance assessment.

Activity 6 — POV, How Might We & Brainstorming

Objective: LO2, LO3 / A1, A2, A3, A4, K7 — convert empathy findings into a POV problem statement, generate a high volume of ideas and select the strongest.

Goal: Take the empathy map from the previous lab and convert it into a sharp Point of View problem statement, reframe it as a set of 'How Might We' questions, then run a timed brainstorm and converge on the best ideas using dot voting.

What you'll produce

A validated POV statement, three HMW questions, 20+ generated ideas and a shortlist of three. (Ed-tools: Design Thinking Toolkit — POV, HMW and Brainstorm canvases (alfredang.github.io/designthinking).)

Step-by-step

1. Open the POV canvas at https://alfredang.github.io/designthinking/.
2. Write your POV using the template: [USER] needs a way to [USER'S NEED] because [INSIGHT].
3. Check the quality: the need must be a verb, and the insight must be a discovery — not a restatement of the need.
4. List any underlying assumptions you are making, so they can be tested later.
5. Reframe the POV into three 'How Might We' questions — broad enough to allow many answers, narrow enough to give direction.
6. Warm-up brainstorm: in 2 minutes, list as many creative uses for a paperclip as you can. Aim for quantity, defer judgement.
7. Run the real brainstorm on your best HMW for 5 minutes. Rules: go for volume, build on others' ideas, one conversation at a time, no criticism, encourage wild ideas.
8. Target at least 20 ideas. Remember: 10 ideas beat 3, and 200 beat 50.
9. Converge — use dot voting or a Now-Wow-How matrix to select the three strongest ideas.
10. Record why each shortlisted idea was chosen, so the rationale survives to the prototype stage.

You're done when

Your POV has a verb-based need and a genuine insight, you generated 20 or more ideas, and you have three shortlisted ideas each with a stated reason for selection.

Note: The full activity brief is in labs/lab-06-*.md. Keep your output — it is the evidence used in the Practical Performance assessment.

Activity 7 — Prototype, Test & Measure the Outcome

Objective: LO5, LO6 / A3, A5, A6, K7 — build a low-fidelity prototype, test it with a user, define the metrics that prove it worked, and communicate the outcome and its value to stakeholders.

Goal: Turn your shortlisted idea into a low-fidelity prototype, test it with a real user, then define how you would measure whether it succeeded — moving from 'we like this idea' to 'here is the evidence it works'. Finish by handing the concept over for delivery with a RACI matrix and a sprint plan.

What you'll produce

A tested prototype, a completed evaluation matrix, an ROI/metrics plan and a RACI handover. (Ed-tools: Design Thinking Toolkit (alfredang.github.io/designthinking), RACI Matrix (alfredang.github.io/raci), Scrum Planner (alfredang.github.io/scrum), BCM (alfredang.github.io/bcm).)

Step-by-step

1. Open the Prototype canvas at https://alfredang.github.io/designthinking/ and pick your top idea from Lab 6.
2. Decide the fidelity: low-fidelity (sketch, storyboard, paper model) is correct at this stage. Just start building — do not polish.
3. Build the prototype in 10 minutes. Remember what you are testing for, and build with the user in mind.
4. Write down your top assumption — the one thing that must be true for this idea to work.
5. Design the cheapest, fastest test that could disprove that assumption. Define where, when and with whom you will run it.
6. Test with a partner: show, don't tell. Let them use it, ask them to think aloud, and note every point of confusion.
7. Record feedback in an evaluation matrix, comparing your prototype against at least one alternative so the user can express a preference.
8. Define your success metrics across four categories: traditional KPIs, customer feedback (CSAT/NPS), design-thinking activity metrics and quick results.
9. Plot your concept on the business-value versus novelty grid — valuable and novel is the target quadrant.
10. Open https://alfredang.github.io/raci/ and assign Responsible, Accountable, Consulted and Informed roles for taking the concept forward.
11. Open https://alfredang.github.io/scrum/ and convert the concept into a prioritised backlog with a first sprint goal.
12. Open https://alfredang.github.io/bcm/ and note one operational risk the solution introduces, plus its mitigation.
13. COMMUNICATE THE OUTCOME — prepare a 2-minute pitch to your stakeholder (the budget holder). State the original problem, the insight you discovered, the prototype, what the test showed, and the metrics that will prove success.
14. Lead with the insight, not the solution — a stakeholder funds a problem they now understand. Contrast your evidence with the obvious solution they would otherwise have paid for.
15. Deliver the pitch to your partner. Ask them: would you fund this? If not, what evidence is missing? Note their answer — that gap is your next test.

You're done when

Your prototype was changed by user feedback, you have named the assumption you tested and the metrics that would prove success, the concept has an owner in the RACI matrix and a first sprint goal, and you have pitched the outcome and its value to your stakeholder leading with the insight rather than the solution.

Note: The full activity brief is in labs/lab-07-*.md. Keep your output — it is the evidence used in the Practical Performance assessment.

Preparing for the Assessment

Oral Questioning (OQ)

- Open book, 20 minutes. The assessor asks you the questions verbally and records your answers — there is no written paper for this instrument.
- Answer in your own words and give an example wherever you can; the assessor may ask a follow-up to confirm your understanding.
- You may refer to these slides, this Learner Guide and approved materials while you answer.
- Revise the definition of design thinking and why it is human-centred (Topic 1).
- Be able to name the traits and mindset of a design thinker, and contrast them with traditional thinking.
- Know the three lenses — desirability, viability, feasibility — and what each asks.
- Be able to describe at least two real use cases and what changed as a result (Topic 2).
- Know all five action phases in order, and the objective of each (Topic 3).
- Know the four quadrants of the empathy map and the POV formula (Topic 4).
- Be able to list metric categories used to measure design outcomes (Topic 4).

Practical Performance (PP)

- Open book, 70 minutes, using the browser-based ed-tools.
- You will be asked to perform design-thinking tasks of the same kind as the class activities.
- Expect to build or interpret an empathy map, write a POV statement, generate and select ideas, describe a prototype and state how you would measure success.
- Your class activity outputs are valid reference material — keep them open.
- Write in full sentences and justify your choices; the assessor is looking for reasoning, not just an answer.

Glossary

- **Design thinking —** A human-centred, iterative problem-solving approach that aims to improve people's experience.
- **Empathize —** The first action phase — understanding the user's experience, situation and emotion by observing, engaging and immersing.

- **Define —** The second action phase — synthesising findings into an actionable problem statement.
- **Ideate —** The third action phase — generating a wide variety and large quantity of possible solutions.
- **Prototype —** The fourth action phase — building a cheap, fast, tangible representation of an idea.
- **Test —** The fifth action phase — putting the prototype in users' hands to gather feedback and refine.
- **Empathy map —** A four-quadrant canvas (Says, Thinks, Does, Feels) capturing a user as a whole.
- **Persona —** A named, fictional but evidence-based representation of a user segment, with goals and frustrations.
- **Journey map —** A chart of every touchpoint a user has with a service over time, exposing pain points.
- **Point of View (POV) —** A problem statement in the form: [USER] needs a way to [NEED] because [INSIGHT].
- **How Might We (HMW) —** A reframing of a POV as an open question broad enough to allow many answers.
- **Divergent thinking —** Deliberately generating many options before evaluating any of them.
- **Convergent thinking —** Narrowing a wide set of options down to the strongest few.
- **Low-fidelity prototype —** A basic, quick model using simple materials — a sketch, storyboard or paper mock-up.
- **High-fidelity prototype —** A prototype close to the finished product, used later to test detail.
- **Desirability / Viability / Feasibility —** The three lenses of organisational design thinking: user desire, business viability, technical feasibility.
- **Beginner's mindset —** Consciously setting aside your own assumptions and expertise when observing users.
- **Net Promoter Score (NPS) —** A customer-feedback metric measuring how likely users are to recommend a product or service.
- **RACI —** A responsibility matrix assigning who is Responsible, Accountable, Consulted and Informed.
- **TRAQOM —** The mandatory SSG training-quality survey learners complete at the end of a WSQ course.

References and Further Reading

- Stanford d.school — An Introduction to Design Thinking Process Guide, dschool.stanford.edu
- Tim Brown, IDEO — Change by Design and ideo.com/case-study
- IBM Design Thinking — ibm.com/design/thinking/

- Design Management Institute — dmi.org/page/2015DVIandOTW (design value index)
- This is Design Thinking — thisisdesignthinking.net/category/cases/
- IDEO Design Kit case studies — designkit.org/case-studies
- Harvard Business Review — 'Better Service, Faster: A Design Thinking Case Study', hbr.org/2016/01
- Nielsen Norman Group — empathy mapping and journey mapping guidance, nngroup.com